

RV UNIVERSITY, BENGALURU

SCHOOL OF COMPUTER SCIENCE AND ENGINEERING

A Project Report On

Mini Fitness Tracker: A Smart Fitness Tracking Mobile Application

B.Tech (Honors)

In

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CERTIFICATE

Certified that the project work titled "**Mini Fitness Tracker: A Smart Fitness Tracking Mobile Application**" is carried out by Bhoomika Kotresh (1RUA24CSE0091), Akhil (1RUA24CSE0028), AliAsghar (1RUA24CSE0035), Chetan B. (1RUA24CSE0103), Chandana M. (1RUA24CSE0098), and Danyatha Y. K. (1RUA24CSE0114) at RV University, Bengaluru, **B.Tech (Hons) in the School of Computer Science and Engineering** of the RV University, Bengaluru during the year 2025-2026. It is certified that all corrections/suggestions indicated for the Internal Assessment have been incorporated in the project report. The Project report has been approved as it satisfies the academic requirements in respect of project work prescribed by the institution.

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Name of Examiners

Signature with Date

- 1.
- 2.

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DECLARATION

We, Bhoomika Kotresh, Akhil, Ali Asghar, Chetan.B, Chandana M, Danyatha Y.K., students of fourth semester B.Tech(Hons), SoCSE, RV University, Bengaluru, hereby declare that the project titled '*Mini Fitness Tracker: A Smart Fitness Tracking Mobile Application*' has been carried out by me and submitted in partial fulfillment of Bachelor of Technology (Hons) in School of Computer Science and Engineering during the year 2024-25.

Further, We declare that the content of the report has not been submitted previously by anybody or to any other university.

We also declare that any Intellectual Property Rights generated out of this project carried out at RV University will be the property of RV University, Bengaluru, and We will be one of the authors of the same.

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Abstract

MiniFitnessTracker is an Android-based mobile application developed to help users monitor and improve their daily fitness activities. The application provides users with features such as step counting, calorie tracking, activity monitoring, progress visualization, profile management, and fitness history tracking.

The application is built using Java/Kotlin in Android Studio with XML-based UI design. Firebase Authentication and Firebase Realtime Database/Firestore are used for secure user login, signup, and data storage. SharedPreferences are used for session management and storing user details locally.

The application tracks user movement using Android activity recognition and step counter sensors. Users can monitor the number of steps walked, calories burned, and distance covered throughout the day. A clean dark-themed user interface with modern fitness visuals improves user experience and engagement.

The project demonstrates the integration of mobile sensors, Firebase services, RecyclerView adapters, SharedPreferences, and modern Android application development practices into a real-world fitness solution.

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Chapter 1: Introduction

1.1 General Introduction:

In today's digital world, maintaining physical fitness has become increasingly important due to sedentary lifestyles and reduced physical activity. Many individuals struggle to track their daily movement, calorie expenditure, and workout consistency.

MiniFitnessTracker is designed as a smart mobile fitness assistant that helps users monitor their health activities in real time. The application allows users to track daily steps, calculate calories burned, measure walking distance, and maintain personal fitness records.

The system aims to motivate users toward healthier habits by providing a simple and interactive interface for activity tracking.

1.2 Literature Survey / Related Work:

Several fitness applications such as Google Fit, Samsung Health, and Fitbit provide activity tracking and health monitoring features. However, many existing systems are either device-dependent, require paid subscriptions, or include complex interfaces that are difficult for beginners.

Research on mobile fitness systems shows that smartphone-based step tracking using built-in sensors provides an efficient and low-cost solution for health monitoring. Studies also emphasize the importance of user-friendly dashboards, progress tracking, and activity visualization for improving user motivation.

MiniFitnessTracker focuses on simplicity, accessibility, and lightweight performance while still offering essential fitness tracking features.

1.3 Problem Statement

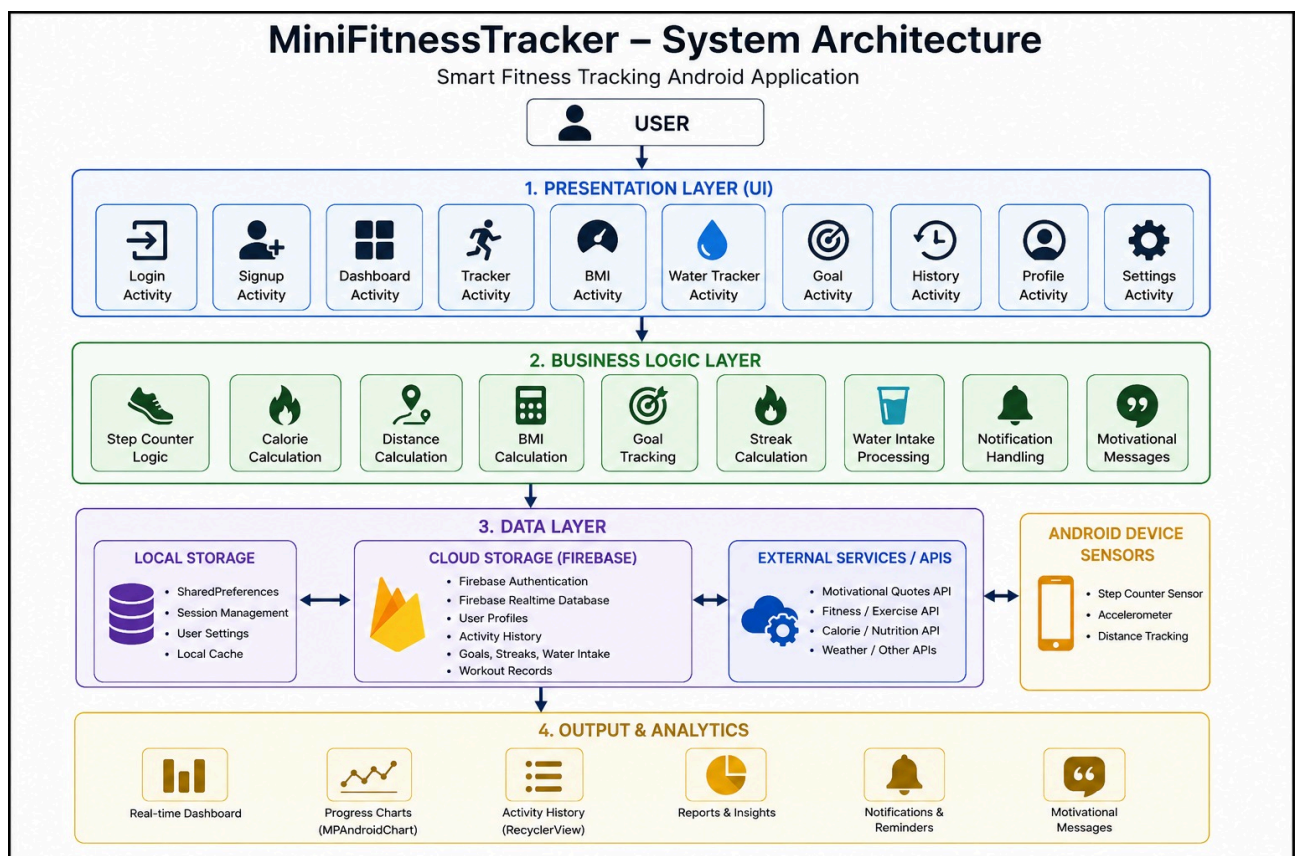
Many users lack an easy-to-use and lightweight mobile application to monitor their daily physical activities such as walking, calorie burn, and movement tracking. Existing applications may require expensive wearable devices, consume excessive resources, or provide overly complex interfaces.

Objectives

1. To develop a mobile-based fitness tracking application.
2. To monitor daily step count using smartphone sensors.
3. To calculate calories burned and distance covered.
4. To provide user authentication and profile management.
5. To maintain activity history and fitness reports.
6. To create an attractive and user-friendly dark-themed interface.
7. To implement Firebase integration for authentication and data storage.

Chapter 2: System Design

2.1 Architectural Design



The application follows a modular Android architecture consisting of:

1. Presentation Layer

- LoginActivity
- SignupActivity
- DashboardActivity
- TrackerActivity
- ProfileActivity
- HistoryActivity
- SettingsActivity

2. Business Logic Layer

Handles:

- Step counting logic
- Calorie calculation
- Distance tracking
- Session handling
- User data processing

3. Data Layer

Uses:

- Firebase Authentication
- Firebase Database / Firestore
- SharedPreferences

The application interacts with Android step counter sensors and stores user information securely in Firebase.

2.2 Tech Stack Overview:

Layer	Technology
Language	Kotlin 2.0.21
IDE	Android Studio
Authentication	Firebase Authentication
Database	Firebase Firestore / Realtime Database
Local Storage	SharedPreferences
UI Components	RecyclerView, CardView
APIs	Android Sensor API
Version Control	GitHub
Minimum SDK	Android 10+
Min SDK / Target SDK	31 / 36

[Figure 2: Tech Stack Table]

Chapter 3: Software Requirements

3.1 Functional Requirements

1. User signup and login functionality.
2. Step counting using device sensors.
3. Real-time activity tracking.
4. Calories burned calculation.
5. Distance covered calculation.
6. User profile management.
7. Activity history and report generation.
8. Firebase data storage and retrieval.
9. Session management using SharedPreferences.
10. Navigation between multiple application screens.

3.2 Non-Functional Requirements

The application should provide accurate step tracking.

1. The user interface should be simple and responsive.
2. The system should securely store user data.
3. The application should work smoothly on Android devices.
4. The system should provide fast loading and low memory usage.

3.3 Hardware Requirements

Component	Requirement
RAM	Minimum 4 GB
Processor	Intel i3 or above
Storage	10 GB free space
Android Device	Android 10+

3.4 Software Requirements

Software	Version
Android Studio	Latest Version
JDK	17 or above
Firebase	Latest
Operating System	Windows 10/11

Chapter 4: Implementation and Testing

4.1 Modules

4.1.1 Authentication Module

The application provides secure login and signup functionality using Firebase Authentication.

Features:

- User registration
- Login validation
- Password authentication
- Session persistence using SharedPreferences

4.1.2 Dashboard Module

The dashboard displays the user's fitness summary including:

- Total steps
- Calories burned
- Distance covered
- Daily activity overview

The dashboard acts as the central navigation page for the application.

4.1.3 Activity Tracker Module

This module uses Android step counter sensors to monitor physical activity.

Features:

1. Real-time step tracking
2. Automatic distance calculation
3. Calorie estimation
4. Pause and resume tracking

4.1.4 Profile Management Module

Users can manage their profile details such as:

1. Name
2. Email
3. Personal information
4. Fitness goals

The profile data is stored securely in Firebase.

4.1.5 History and Report Module

The application maintains activity history for users.

Features:

1. Daily activity logs
2. Workout history
3. Progress visualization
4. Fitness reports

RecyclerView and adapters are used to display activity records dynamically..

4.1.6 Settings Module

The settings screen allows users to:

- Manage account settings
- Update preferences
- Logout securely

4.2 Testing Strategies

- **Unit Testing**
 - Individual modules such as login validation, step calculation, and activity tracking were tested independently.
- **Integration Testing**
 - Firebase integration, sensor APIs, and SharedPreferences functionality were tested together.
- **System Testing**
 - Complete application workflow from signup to activity tracking was verified.
- **User Acceptance Testing**
 - The application was tested by peers for usability, interface quality, and performance.

Chapter 5: Results and Discussion

.5.1 System Performance

The application successfully tracked user steps and displayed real-time activity updates.

- Step counter updates were responsive.
- Firebase authentication worked securely.
- Activity reports were generated correctly.
- RecyclerView-based history loaded efficiently.

5.2 User Experience and Feedback

Users appreciated:

- Dark-themed modern interface
- Easy navigation
- Real-time activity tracking
- Attractive dashboard design
- Lightweight performance

The application provided a beginner-friendly experience suitable for daily fitness monitoring.

5.3 Discussion

The project demonstrates how Android sensors and Firebase services can be integrated into a practical health monitoring system.

The modular architecture improves scalability and maintainability. The use of SharedPreferences enables efficient session handling, while Firebase ensures secure cloud-based storage. The project also highlights the importance of mobile applications in promoting healthier lifestyles.

5.4 Summary

MiniFitnessTracker successfully achieved its objectives by:

1. Tracking daily steps accurately.
2. Calculating calories and distance.
3. Providing secure authentication.
4. Maintaining user fitness history.
5. Delivering a clean and interactive UI.

Chapter 6: Conclusion and Future Work

MiniFitnessTracker is a smart Android fitness application developed to encourage users to maintain healthy habits through activity monitoring and progress tracking. The application integrates Android sensors, Firebase services, RecyclerView, adapters, SharedPreferences, and modern Android UI design to provide an efficient fitness management solution. The project successfully demonstrates practical Android development concepts and real-world application implementation.

Future Enhancements

1. Integration with wearable fitness devices.
2. AI-based workout recommendations.
3. Diet and nutrition tracking.
4. Cloud backup and synchronization.
5. Social fitness challenges and leaderboards.
6. Push notifications and fitness reminders.
7. Integration with Google Fit APIs.

REFERENCES

1. Android Developers Documentation – <https://developer.android.com>
2. Firebase Documentation – <https://firebase.google.com>
3. Google Fit API Documentation – <https://developers.google.com/fit>
4. W3Schools Android Tutorials – <https://www.w3schools.com>
5. Android Sensor API Guide – <https://developer.android.com/guide/topics/sensors>

Appendices

Appendix 1: Screenshots

LOGIN SCREEN

Login



Welcome

Login to continue your fitness journey

[Forgot Password?](#)

Login

or

Create New Account

Welcome 🖐️

Keep pushing your limits!

Main Features

AI Fitness Buddy 🤖

Track Activity 🔥

Water Intake 💧

BMI Calculator ⚖️

Sleep Calculator 😴

Diet Plan 🍏

Meditation 🧘

Sleep Tracker 🌙

14:19



BMI Calculator

Calculate your Body Mass Index

Weight (kg)

Height (cm)

CALCULATE BMI

Result: --

BACK



AI CHATBOT SCREEN:

 **AI Fitness Buddy**

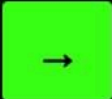
Hello! I'm your AI Fitness Buddy.
Ask me anything about workouts,
diet, or health!

Try asking:

Best diet for weight loss?

How to gain muscle fast?

Ask anything...



History

Your past workout records

Recent Activity

No history available for yesterday

[Back to Dashboard](#)

14:19

93%

Diet Plan

Optimized nutrition for your goals

Suggested Meals

- Breakfast: Oatmeal with berries
- Lunch: Grilled chicken salad
- Snack: Greek yogurt
- Dinner: Baked salmon with broccoli

REFRESH DIET PLAN

BACK



14:19

93%



Sleep Calculator 🧠💤

Wake up refreshed by sleeping in cycles

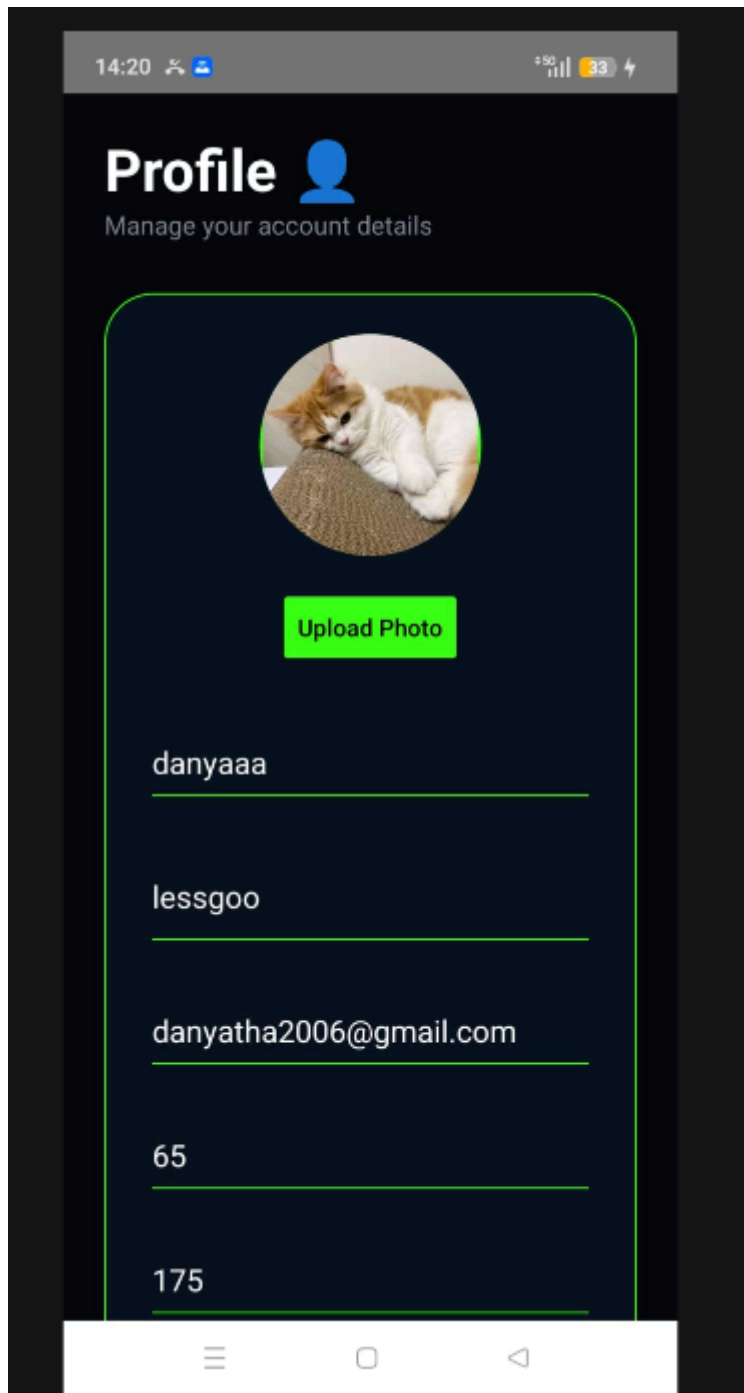
If I go to bed now, I should wake up at:

CALCULATE WAKE UP TIMES

BACK



KYC Verification Screen:



Appendix 2: Source Code

All source code is available at the following GitHub repository: Microsoft.QuickAction.Bluetooth:
<https://github.com/Danyatha2006/MiniFitnessTracker>

Appendix 3: Plagiarism Details

The report was verified for originality using a plagiarism checker. The majority of the content is original work produced by the author. Technical references to official documentation (Firebase, Google) account for the small percentage of matched content. The plagiarism report is available at:

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